AQUASHIELD CPS — HOW IT WORKS

Plain English Breakdown: How the Computer Solves Every Attack Automatically

Sustainathon 2024 • VIT Vellore
Zero Jargon • Simple Everyday Language Guide

❓ The Big Question: Does a human fix it, or does the computer fix it by itself?

Answer: The computer fixes the emergency 100% BY ITSELF automatically in less than 2 seconds!
Humans do NOT run around with screwdrivers during an attack. Think of an airplane on autopilot: if a sensor breaks mid-flight, the autopilot doesn't crash the plane and wait 2 hours for a mechanic on the ground. It immediately uses backup math and keeps flying safely.
In a water plant, waiting for a human means **thousands of liters of clean water will already be flooded, wasted, or poisoned**. AquaShield saves the plant instantly by itself, and only sends a summary email to humans later.

How AquaShield Solves Each of the 5 Problems Step-by-Step

Problem 1: The Sensor Lies

Layer 1: The Eyes

What the hacker did: Hacked the water level sensor on Tank 2 to freeze at **32%**, making the computer think the tank is almost empty.

Why old security tools failed: They blindly trust the sensor number. They think the tank is drying up, pump in way too much water, and **flood the whole building**.

How AquaShield solves it automatically: *The "Virtual Sensor" (Math Backup)*
1. AquaShield checks the water flow: "Wait! Water is pouring in at 42 Liters every second, and only 38 Liters are leaving. By simple math, the tank MUST be at 73%!"
2. AquaShield mutes the lying sensor.
3. Instead of reading the broken sensor, the computer uses its own live calculated math estimate.
4. The plant keeps pumping clean water with **zero seconds of downtime**!

**What humans do later:* A technician casually comes next week to replace the plastic sensor probe.*

Problem 2: The Network Cable is Jammed

Layer 2: The Nerves

What the hacker did: Trapped messages inside the network wire, delaying them by 385 milliseconds and resending old data.

Why old security tools failed: They see messages arriving late, panic with a "NETWORK TIMEOUT" error, and shut down the whole plant, dumping 40,000 Liters of water into the sewer.

How AquaShield solves it automatically: *Automatic Wire Switch (Failover)*
1. AquaShield notices the water tanks and pumps are completely physically healthy, but messages on that specific wire are late.
2. In 38 milliseconds, AquaShield automatically flips a software switch to a secondary backup wireless channel.
3. Messages immediately start arriving on time again. The plant never stops.

**What humans do later:* An IT person checks the cable firewall during scheduled maintenance.*

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Problems 3, 4, 5 & The 15-Second Judge Speech

Problem 3: The Computer Brain is Hacked

Layer 3: The Brain

What the hacker did: Broke into the factory computer and told it to push water pressure up to a dangerous **8.5 bar** (which will tear apart the water filter membranes).

Why old security tools failed: The computer thinks: "An operator told me to do this, so I will do it." It doesn't know the command is dangerous.

How AquaShield solves it automatically: *Hardcoded Safety Clamp*

1. AquaShield has an unbreakable physical rule: “The plastic water filters tear apart at 5.0 bar. No command is EVER allowed to exceed 4.0 bar.”
2. The moment the hacked code tries to push past the limit, AquaShield automatically clamps the pressure back down to a safe 3.2 bar.
3. The hacker's command is blocked at the gate. The filters and pipes are saved.

**What humans do later:* Security resets the factory computer's password.*

Problem 4: Bad Software Code (Stuxnet)

Layer 4: The Instincts

What the hacker did: Replaced the pump motor’s code with bad code that violently shakes the motor to destroy it, while **lying to the screen** that the motor is running at a normal 1750 RPM.

Why old security tools failed: They only look at the screen. The screen says "Everything is fine!", so nobody notices the motor shaking until it catches fire.

How AquaShield solves it automatically: *Digital Fingerprint Check & Auto-Restore*

1. Every piece of software code has a unique "digital fingerprint" (hash).
2. AquaShield checks the motor's code fingerprint. It sees that the fingerprint doesn't match the original factory code!
3. In 0.4 seconds, AquaShield erases the hacked code and restores the clean, read-only backup software.
4. The motor stops vibrating and returns to smooth operation.

**What humans do later:* Nothing! The system already cleaned and restored itself.*

Problem 5: A Valve is Physically Jammed Open

Layer 5: The Muscles

What the hacker did: A mechanical release valve physically jammed open at 100%, and clean water is gushing out at 65 Liters/second.

Why old security tools failed: The computer told the valve to "CLOSE". Old tools assume the valve obeyed, and blame the water sensor for "glitching". Meanwhile, 60,000 Liters of water leak away.

How AquaShield solves it automatically: *Backup Pipe Diverter*

1. AquaShield compares the computer command (0% Closed) with the flow meter (65 Liters/sec leaking).
2. The software and sensors are healthy, so the metal valve must be physically stuck.
3. AquaShield automatically opens a secondary motorized bypass valve (Valve 202B) and lowers pump speed to stop the leak instantly.
4. Clean water is safely routed through the backup pipe.

**What humans do later:* A maintenance worker goes to physically unjam the valve with a wrench.*

🎤 Your 15-Second Answer for Judges Tomorrow

“In critical water plants, humans cannot react in milliseconds. If you wait for a human, thousands of liters of clean water are already lost.

***AquaShield is 100% autonomous:** it detects the breach, identifies whether it’s a sensor, cable, computer, software, or valve, and **deploys a software backup in 38 milliseconds** without dropping water production. Humans are only notified afterward for routine maintenance.”*